

Week 5 Summary Sheet

Wanrong Yang

24/10/2024

Pumping Lemma for Regular Languages

1. Introduction

- not every language is regular language
- it is for proving a certain language is **not regular**

2. Theorem

- let L be a regular language (it can be accepted by a DFA)
- then there exists a constant n , such that for every string w in L , $|w| \geq n$
- n means the number of states of the DFA
- We can break w into 3 strings, $w = xyz$ such that
 1. $y \neq \varepsilon$ or $|y| > 0$
 2. $|xy| \leq n$
 3. For all $k \geq 0$, the string xy^kz is also in L
- if there is any contradiction, then the language is not regular

3. Example

- prove that $L = \{0^n 1^n | n \geq 1\}$ is not regular
- **Step 1** Assume L is regular
- **Step 2** Let n be a constant, the number of states of DFA
- **Step 3** Let $w = 0^n 1^n$
- **Step 4** Split $w = xyz$ such that
 - $y \neq \varepsilon$
 - $|xy| \leq n$
 - for all $k \geq 0$, the string xy^kz is also in L

Find contradiction to proof.

4. Finite language is always a regular language